

NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY
FACULTY OF DATA SCIENCE & ARTIFICIAL INTELLIGENCE (FDA)

OFFICIAL MIDTERM EXAMINATION
Academic Year: 2026–2027
Semester: Fall 2026 Course Code:
DSAI1003

FUNDAMENTAL PROGRAMMING CONCEPTS IN PYTHON

PROGRESS EXAM: NUMBERS, STRINGS, AND OPERATORS

Paper Code: _____ | Duration: 60 Minutes (Excluding paper distribution)

Candidate Full Name:

Student ID (MSSV): ...

Class / Cohort:

Exam Room / Seat No.:

Section	A	B	C	D	Total (100)	Examiner Signature
Max Score	20	25	25	30	100	
Score						

Instructions: Closed-book examination. All electronic devices and internet access are prohibited. Answer all questions directly in the designated spaces. In programming tasks, use the **Input** → **Process** → **Output** pattern. Do **not** define user functions (def) or use loops/branching. Assume users enter data in the required format. Do not detach pages.

MULTIPLE-CHOICE ANSWER SHEET (Write A, B, C, or D in the corresponding box):

Question	1	2	3	4	5	6	7	8	9	10
Answer										

Section A: Numbers, Strings & Operators Concepts

(20 points)

Answer all 10 questions. Each correct answer carries 2 points. Suggested time: 12 minutes.

- Q1.** Which of the following is a **valid** variable name according to Python identifier rules?
- 2nd_total
 - tax-rate
 - _unit_price
 - class
- Q2.** What is the precise effect of executing the assignment statement `count = count + 1`?
- It checks whether count is mathematically equal to `count + 1`.
 - It evaluates `count + 1` using the current value of count and assigns the result back to count.
 - It raises a syntax error because a variable cannot appear on both sides of `=`.
 - It permanently converts the data type of count into a floating-point number.
- Q3.** What value and data type are produced by the floor division expression `19 // 4` in Python?
- 4.75 (type float)
 - 4 (type int)
 - 3 (type int)
 - 4.0 (type float)
- Q4.** Following standard operator precedence, what is the result of the expression `2 + 3 * 2 ** 3`?
- 26
 - 40
 - 200
 - 216
- Q5.** For any positive integer n, which expression correctly extracts its **last digit** (units digit)?
- `n // 10`
 - `n / 10`
 - `n % 10`
 - `n % 100`

- Q6.** Which statement regarding mathematical functions and the math module in Python is correct?
- A. The `sqrt()` function is a built-in function that does not require an import statement.
 - B. Built-in functions like `abs()` and `round()` can be called directly without importing math.
 - C. The constant `math.pi` is a built-in keyword in Python available in all scopes.
 - D. The math module cannot be used with floating-point numbers.
- Q7.** Why can the comparison `0.1 + 0.2 == 0.3` evaluate to False in Python?
- A. Python forbids addition between decimal floating-point values.
 - B. Binary floating-point representation causes slight roundoff errors for certain decimal fractions.
 - C. The `==` operator cannot compare floating-point operands.
 - D. Python automatically converts `0.1 + 0.2` into an integer before comparison.
- Q8.** Given the string assignment `s = "PYTHON"`, what are the values of `s[1]` and `s[-1]`?
- A. 'P' and 'N'
 - B. 'Y' and 'N'
 - C. 'Y' and 'O'
 - D. 'P' and 'O'
- Q9.** What is the exact result of the string expression `"Data" + "5" * 2`?
- A. "Data10"
 - B. "Data55"
 - C. "Data5Data5"
 - D. A `TypeError` occurs because strings cannot be multiplied by integers.
- Q10.** Consider the code: `word = "python"; word.upper(); print(word)`. What is printed?
- A. PYTHON
 - B. python
 - C. Python
 - D. An `AttributeError` occurs because strings do not possess an `upper()` method.

Section B: Output Tracing, Type Conversion & Error Diagnosis

(25 points)

Answer all 5 questions. Each question carries 5 points. Suggested time: 15 minutes.

Q11. Exact Output Tracing with Arithmetic & Modulo (5 points)

Predict the **exact screen output**, including spaces, punctuation, and line breaks:

```
a = 23
b = 5
quotient = a // b
remainder = a % b
power_val = b ** 2 - a
print("q:", quotient, "r:", remainder, sep=" | ", end="\n")
print(f"Check: {quotient * b + remainder == a}")
print(f"Power diff: {power_val}")
```

Output Box for Q11

Q12. String Operations, Indexing and Length Properties (5 points)

Consider the following Python program:

```
title = "Computer"
n = len(title)
first = title[0]
mid = title[n // 2]
last = title[-1]
combo = (first + mid + last).lower()
print(f"n = {n}")
print(f"Parts: {first}-{mid}-{last}")
print(f"Result: {combo * 2}")
```

- Output of `print(f"n = {n}")`: _____
- Output of `print(f"Parts: {first}-{mid}-{last}")`: _____
- Output of `print(f"Result: {combo * 2}")`: _____
- Explain why integer floor division (`n // 2`) is required rather than `n / 2` for the index: _____

Q13. Diagnose and Correct Common Type and Operator Errors (5 points)

For each case, state the error type / problem and write a corrected Python statement.

Code / Situation	Problem / Error	Correction / Valid Code
radius = input("Radius: ") area = 3.14159 * radius ** 2		
s = "Python" s[0] = "J" Goal: change first letter to 'J'		
total = 150 msg = "Total is: " + total		

Q14. Formula Evaluation with the math Module (5 points)

Predict the exact output of the following geometry calculation:

```
import math

a = 3.0
b = 4.0
hypotenuse = math.sqrt(a ** 2 + b ** 2)
area = 0.5 * a * b
print(f"Sides: {a:.1f}, {b:.1f}")
print(f"Hypotenuse: {hypotenuse:.2f}")
print(f"Triangle Area: {area:.1f}")
```

Output Box for Q14

Q15. Replacing Magic Numbers with Named Constants (5 points)

Consider the following soda volume calculation:

```
cans = 6
bottles = 2
total_liters = cans * 0.355 + bottles * 2.0
```

- State the two **magic numbers** present in line 3: _____
- What is the Python naming convention (PEP 8) for named constants? _____
- Rewrite line 3 using well-named constants `CAN_VOLUME` and `BOTTLE_VOLUME`:

Section C: Program Design Using the Input → Process → Output Model

(25 points)

*Answer all tasks. Suggested time: 15 minutes.***Problem Specification: Cash Change Breakdown Calculator (Dollars, Quarters, Cents)**

A cash register program receives an integer representing the total change due to a customer in **cents** (e.g., 387). It computes the maximum number of full dollars (\$1 = 100 cents), the number of 25-cent quarters from the remainder, and the leftover cents. It also displays the total amount formatted as dollars and cents.

Task 1: Identify the IPO Components (6 points)

- **Input(s):** _____
- **Process:** _____
- **Output:** _____

Task 2: Write the Program Logic in Ordered Steps (7 points)

Write clear sequential steps using the Input → Process → Output structure. Your steps should state the required type conversion, integer floor division (`//`), modulo operations (`%`), and output formatting.

Program-Logic Work Area**Task 3: Hand Trace (6 points)**

For `total_cents = 387`, show the step-by-step arithmetic calculations and the exact formatted result.

Hand-Trace Calculation**Task 4: Explain Two Design Choices (6 points)**

- a) Why must the user input be converted with `int()` rather than `float()` for this algorithm?
- b) Why are floor division (`//`) and remainder (`%`) used instead of true division (`/`) when finding coin quantities?

Section D: Applied Programming in Python

(30 points)

Write complete, syntactically correct Python statements using the **Input** → **Process** → **Output** pattern. Do **not** define custom functions (def), use loops, or use branching. Suggested time: 18 minutes.

Problem 1: Cylindrical Storage Tank Volume and Coating Cost (15 points)

Write a Python program that computes the geometry and protective coating cost for an industrial cylindrical storage tank.

Requirements:

1. Prompt the user for the tank_id (string) using input().
2. Prompt for the tank radius radius (r) in meters and convert it to float.
3. Prompt for the tank height height (h) in meters and convert it to float.
4. Import math to access math.pi. Define a constant COATING_COST_PER_SQM = 14.50.
5. Compute the storage volume: $V = \pi r^2 h$.
6. Compute the total surface area (top, base, and side wall): $A = 2\pi r h + 2\pi r^2$.
7. Compute the total coating cost: total_cost = $A \times \text{COATING_COST_PER_SQM}$.
8. Display a formatted summary using f-strings: show radius and height with 2 decimal places, volume with 3 decimal places, surface_area with 2 decimal places, and total_cost with 2 decimal places prefixed by \$.

The output should follow this general form:

```
Tank ID: TANK-A1
Dimensions: Radius = 2.50 m, Height = 6.00 m
Storage Volume: 117.810 m^3
Total Surface Area: 133.52 m^2
Total Coating Cost: $1936.01
```

Python Solution for Problem 1

Problem 2: Customer Voucher and Masked Account Statement (15 points)

Write a Python program that processes customer data and formats a secure account voucher using string operations and numeric formatting.

Requirements:

1. Prompt the user for the customer's `full_name` (string) using `input()`.
 2. Prompt for an 8-character string `account_code` (e.g., ACCT8492).
 3. Prompt for the current `account_balance` in dollars and convert it to `float`.
 4. Define a named constant `DIVIDER_LENGTH = 42`.
 5. Process the data:
 - Convert `full_name` to uppercase using `.upper()` and obtain its length using `len()`.
 - Extract the first initial: first character of `full_name` in uppercase.
 - Construct a masked account code showing the first two characters, four asterisks ("****"), and the last two characters using string indexing, repetition, and concatenation.
 - Create divider lines using string repetition: `"=" * DIVIDER_LENGTH` and `"-" * DIVIDER_LENGTH`.
 6. Display the voucher formatted exactly as shown below, with `account_balance` formatted to 2 decimal places.
- For example, if the user enters Margaret Hamilton, ACCT8492, and 1450.8, the program outputs:

```
=====
OFFICIAL CUSTOMER ACCOUNT VOUCHER
-----
Customer Name: MARGARET HAMILTON (17 chars)
Initial & Masked Code: M | AC****92
Current Balance: $1450.80
=====
```

Python Solution for Problem 2

— END OF EXAM PAPER —